Chapter 4: Randomized Blocks, Latin Squares, And Related Design

In order to eliminate as much of the natural variation as possible and increase the sensitivity of experiment, it would be advisable to choose the experimental units for a study as **homogeneous** as possible. On the other hand, most experimenters would like the conclusions of their work to have **wide** applicability. Blocking can be used to achieve both objectives.

Blocking is an **extremely important** design technique, used **extensively** in industrial experimentation. We will discuss about blocking in this chapter. In particular, we will discuss following designs:

- Randomized Complete Block Designs (RCBD)
- Latin Square Designs (LSD) and Graeco – Latin squares
- Balanced Incomplete Block Designs (BIBD)

In many experiment, it is impossible to run all experiments under identical/homogeneous conditions, especially if the experiments take a long time. In this case, variability arising from a **nuisance factor** can impact the results.

A nuisance factor is a factor that probably has an effect on the response, but we are not interested in that effect.

Sometimes a nuisance factor is **unknown** and **uncontrolled**; that is we do not know the factor exists and it may even be changing levels while we are conducting the experiment. **Randomization** is the design technique used to guard against such a "**confounding**" or "**lurking**" nuisance factor.

When the nuisance source is **known** and **controllable**, then a design technique called **blocking** can be used to systematically eliminate its effect on the statistical comparisons among treatments.

In a **randomized block design**, a group of **homogeneous** experimental units is used so that the conclusion can be more general; however, these heterogeneous experimental units are grouped into homogeneous subgroups before they are randomly assigned to treatment factor levels. In a randomized block experiment, the treatments are randomized within the blocks.

The act of grouping the experimental units together in a **homogeneous** groups is called **blocking**. Randomly assigning treatment factor levels to experimental units within the smaller homogeneous subgroups of experimental units or blocks has the same effect as using only homogeneous units, yet it allows the conclusion to be generalized to the entire class of heterogeneous experimental units used in the study.

Example 1: Suppose we would like to compare four treatments A, B, C and D in a **randomized complete block design** (**RCBD**) by using five blocks, B1 to B5.

Complete means that each block contains **all** of the treatments.

The following table is an example of RCBD. The numbers in parentheses indicate the random order in which the experiment were run **within** each block.

Block	A	B	C	D
B1	(1)	(3)	(2)	(4)
B2	(4)	(2)	(3)	(1)
B3	(2)	(1)	(4)	(3)
B4	(1)	(3)	(2)	(4)
B5	(3)	(4)	(1)	(2)

R Codes to Create an RCBD:

```
> Block = factor(rep(c("B1","B2","B3","B4","B5"), each=4))

> f  = factor(c("A","B", "C", "D"))

>Treat = c(sample(f,4), sample(f,4), sample(f,4),
sample(f,4),sample(f,4))


>plan<-data.frame(Block= Block, treatment = Treat)

>write.table(plan, file = "RCBPlan.csv", sep= ", ", row.names =
FALSE)
```

An RCBD can also be easily created using user written packages. First use install.packages("agricolae") to install agricolae package. Then use function desig.rcbd to generate an RCBD.

```
>  library(agricolae)
> Treat = c("A","B", "C", "D")
> Outdesign = design.rcbd(Treat, 5, seed=11)

> rcb = Outdesign$book

> levels(rcb$block) = c("B1","B2","B3","B4","B5")
```

You can also use, for example, LSD.test, HSD.test, scheffe.test and SNK.test to do Fisher's LSD test, Tukey's HSD test, Scheffe's test and SNK test, respectively.

Example 2: Consider a randomized complete block design (Narayanan and Adoriso, 1(983), Model studies on plate girders, *Journal of Strain Analysis*, **18**, 111-117) to compare four methods for predicting the shear strength for steel plate girders. Data for nine girders in the form of the ratio of predicted to observed load for these procedures are given in the following table.

Treatments (Methods)

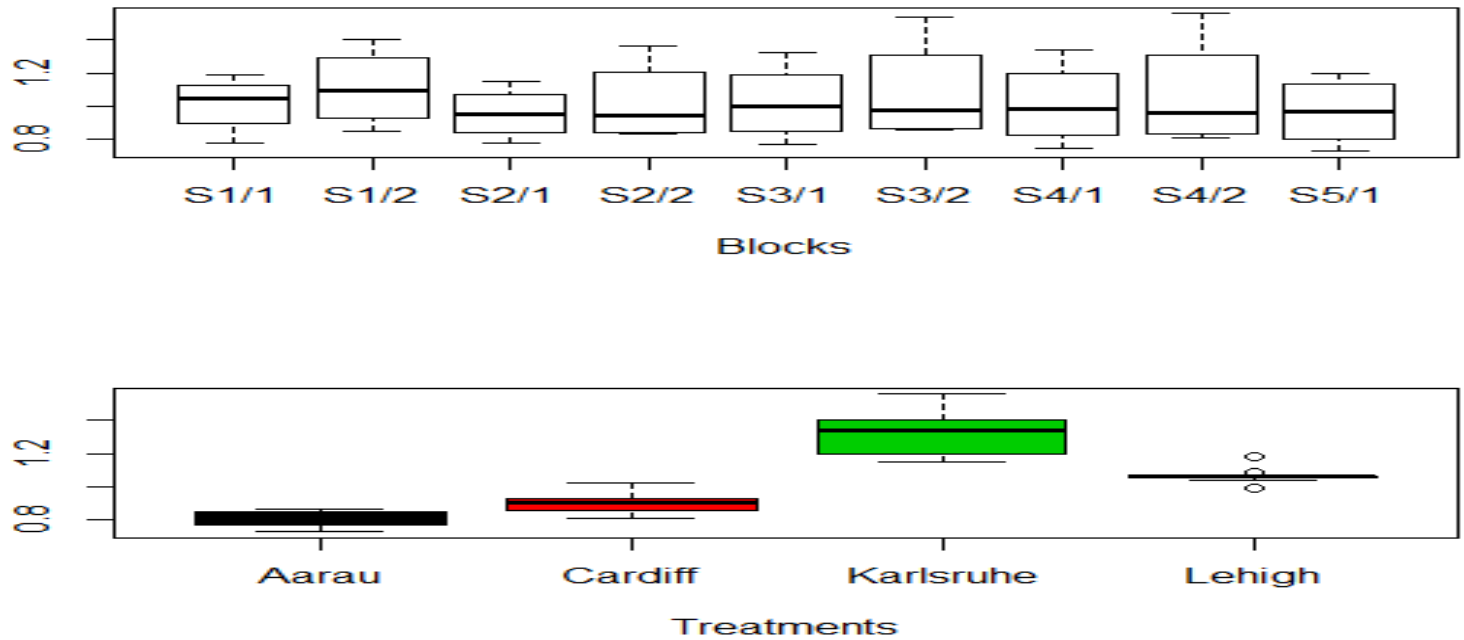
Block (Girder)	Aarau	Karlsruhe	Lehigh	Cardiff
S1/1	0.772	1.186	1.061	1.025
S2/1	0.744	1.151	0.992	0.905
S3/1	0.767	1.322	1.063	0.930
S4/1	0.745	1.339	1.062	0.899
S5/1	0.725	1.200	1.065	0.871
S1/2	0.844	1.402	1.178	1.004
S2/2	0.831	1.365	1.037	0.853
S3/2	0.867	1.537	1.086	0.858
S4/2	0.859	1.599	1.052	0.80

R Codes to Read Data from a File and Generate Block Index and Treatment index:

```
> data = read.csv("Data2_4.csv")

> block = as.factor(rep(data[,1], each =4))

> treat = as.factor(rep(colnames(data)[-1], 9))

> y=c(t(data[,2:5]))
```

R Codes to Obtain Box-Plots:

```
> par(mfrow=c(2,1))
> boxplot(y~block, xlab="Blocks")
> boxplot(y~treat, xlab="Treatments", col=c(1,2,3,4))
```



Statistical Model: For $i = 1,2,\ldots,k$ and $j = 1,2,\ldots,b$, let

Response Variable

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$$

Random Error

Overall Mean

Treatment Effect

Block Effect

Estimates:

- $\hat{\mu} = \bar{y}_{..}$
- $\hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}$ for $i = 1,2,\ldots,k$
- $\hat{\beta}_j = \bar{y}_{.j} - \bar{y}_{..}$ for j= $1,2,\ldots,b$

Then Residuals are:

$$e_{ij} = y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}$$

Statistical Model: For $i = 1,2,\ldots,k$ and $j = 1,2,\ldots,b$, let

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$$

Define

$$SS_{Total} = \sum_{i=1}^{k}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{..})^2 ,$$

$$SS_{Blocks} = \sum_{i=1}^{k}\sum_{j=1}^{b}(\bar{y}_{.j} - \bar{y}_{..})^2 ,$$

$$SS_{Treatments} = \sum_{i=1}^{k}\sum_{j=1}^{b}(\bar{y}_{i.} - \bar{y}_{..})^2$$

$$SS_{Error} = \sum_{i=1}^{k}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})^2$$

It can be shown that

$$SS_{Total} = SS_{Treatments} + SS_{Blocks} + SS_{Error}$$

It can also be shown that

$$E\left(MS_{\text{Treatments}}\right) = \sigma^2 + \frac{b\sum_{i=1}^{k}\tau_i^2}{k-1}$$

$$E\left(MS_{\text{Blocks}}\right) = \sigma^2 + \frac{k\sum_{j=1}^{b}\beta_j^2}{b-1}$$

and

$$E(MS_{Error}) = \sigma^2$$

The **ANOVA table**:

Source	df	SS	MS	F
Treatments	$k-1$	SSTr	$\mathrm{MSTr} = \frac{\mathrm{SSTr}}{\mathrm{k}-1}$	$\mathrm{F} = \frac{\mathrm{MSTr}}{\mathrm{MSE}}$
Blocks	b−1	SSB	$\mathrm{MSB} = \frac{\mathrm{SS}B}{\mathrm{b}-1}$	$\mathrm{F} = \frac{\mathrm{MS}B}{\mathrm{MSE}}$
Error	$(k-1)(\mathrm{b}-1)$	SSE	$\mathrm{MSE} = \frac{\mathrm{SSE}}{(k-1)(b-1)}$	
Total	n−1	SST		

R Codes to Obtain ANOVA Table for RCBD:

```
> model <- lm(y~ block+treat)
> anova(model)
```

Analysis of Variance Table

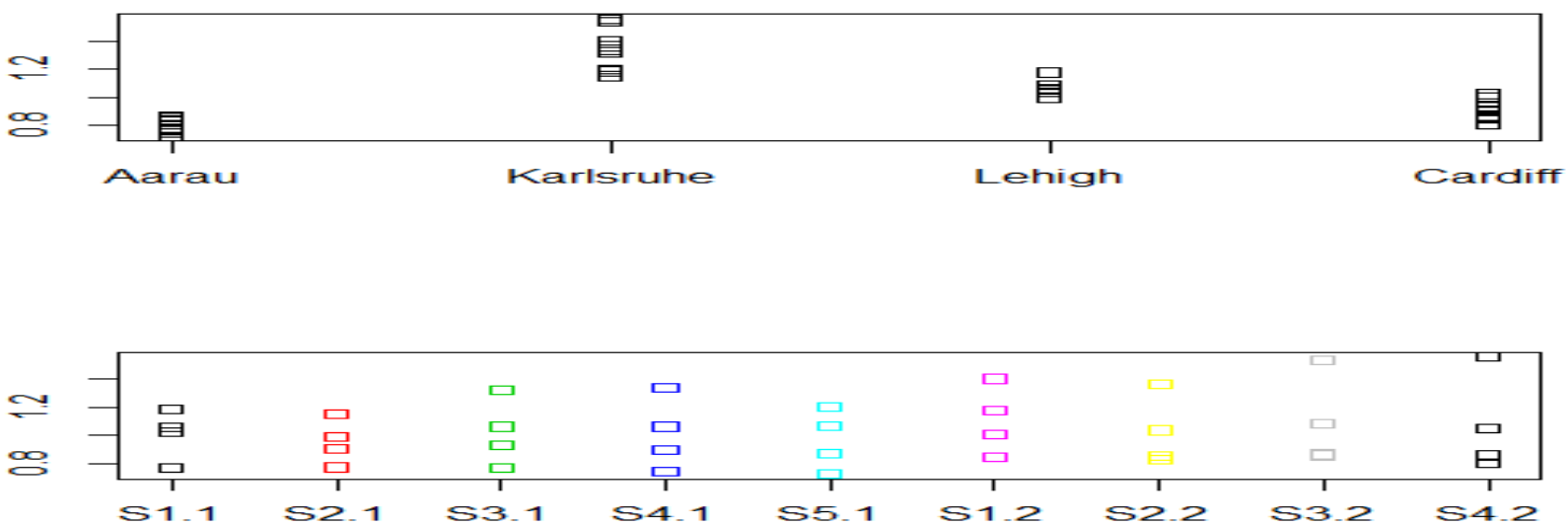
Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
block	8	0.08497	0.01062	1.5164	0.2036
treat	3	1.49997	0.49999	71.3830	4.211e-12 ***
Residuals	24	0.16810	0.00700		

Signif. codes: 0 '*' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1**

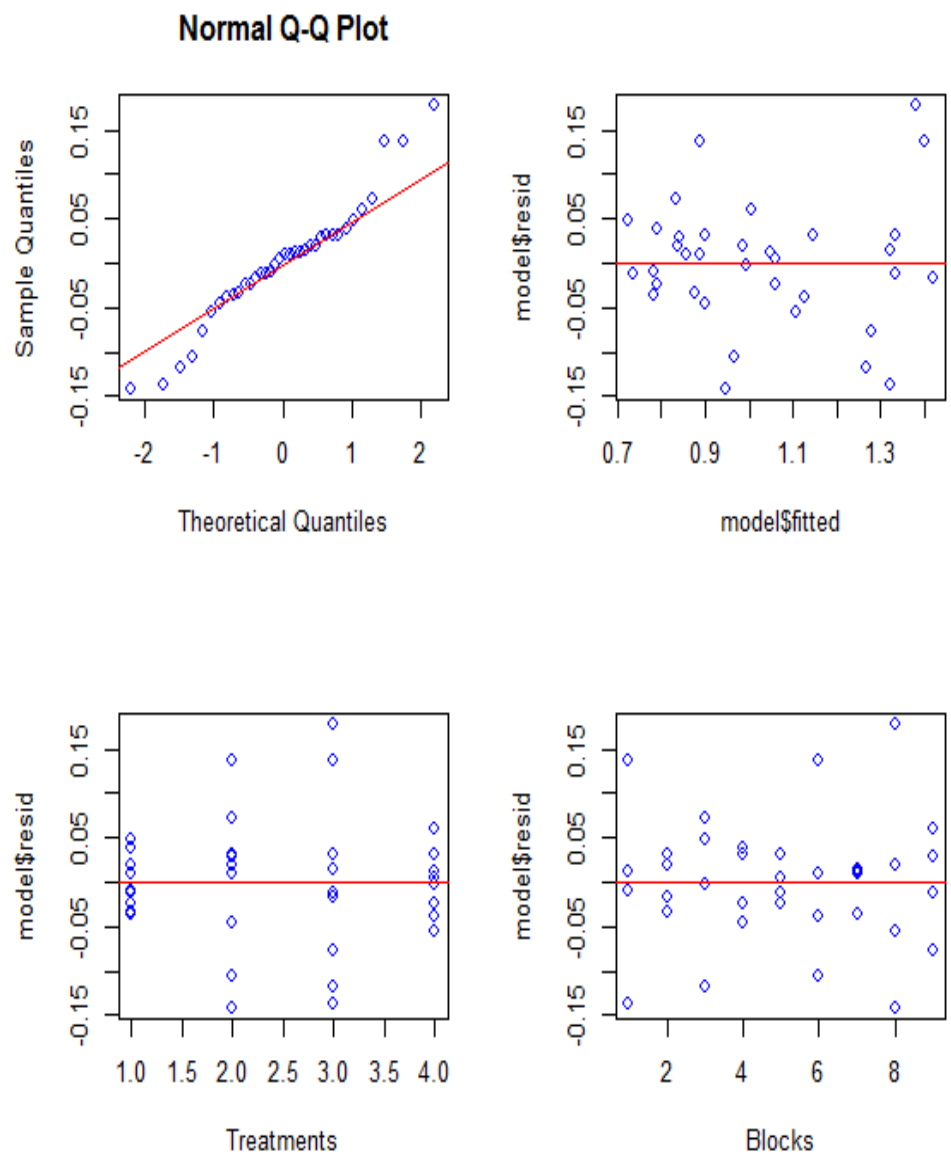
R Codes to Draw Scatter-Plot of Data:

```
> data = read.csv("Data2_4.csv")
> data_sub = data[,2:5]
> rownames(data_sub) = data[,1]
> par(mfrow=c(2,1))
> stripchart(data_sub, vertical =TRUE)
> stripchart(data.frame(t(data_sub)), vertical =TRUE, col=c(1:9))
```



R Codes to Check Assumptions:

```
> windows()
> par(mfrow=c(2,2))
> qqnorm(model$resid,
    col="blue")
> qqline(model$resid, col=2)
> plot(model$fitted,model$resid,
    col="blue")
> abline(h=0, col="red")
> plot(c(treat), model$resid,
    xlab="Treatments",col="blue")
> abline(h=0, col="red")
>plot(c(block), model$resid,
    xlab="Blocks", col="blue")
>abline(h=0, col="red")
```



R Codes to Perform Bartlett's Test :

```
> model = lm(y~ block+treat)
> bartlett.test(model$resid, treat)
```

Bartlett test of homogeneity of variances
data: model$resid and treat
Bartlett's K-squared = 15.2701, df = 3, p-value = 0.0016

```
> bartlett.test(model$resid, block)
```

Bartlett test of homogeneity of variances

data: model$resid and block
Bartlett's K-squared = 15.9999, df = 8, p-value = 0.04238

R Codes to Perform Levene's Test to for Equality of Variances among Treatments:

```
> m.treat=apply(data_sub, 2, median)
> diff <- abs(y-rep(m, times=9))
> model.d.treat <- lm(diff~treat)
> anova(model.d.treat)
```

Analysis of Variance Table

Response: diff

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treat	3	0.036103	0.0120343	3.9452	0.01677 *
Residuals	32	0.097612	0.0030504		

Signif. codes: 0 '*' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1**

R Codes to Perform Levene's Test to for Equality of Variances among Blocks:

```
> m.block=apply(data_sub, 1, median)
> diff <- abs(y-rep(m.block, each=4))
> model.d.block <- lm(diff~treat)
> anova(model.d.block)
```

Analysis of Variance Table

Response: diff

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
block	8	0.05601	0.0070016	0.297	0.9607
Residuals	27	0.63649	0.0235736		

Non-Parametric Test: The analogue of the Kruskal-Wallis test, for a RCBD, is **Friedman's test**.

Friedman's test can be used for analyzing **un-replicated complete block designs** (i.e., there is exactly one observation in response variable for each combination of levels of groups and blocks) where the **normality** assumption may be violated.

In Friedman's test , the observations are replaced by their **ranks within** each block, and the usual ANOVA is run.

```
> data.ranks= apply(data_sub,1,rank)
> data.ranks
```

	S1/1	S2/1	S3/1	S4/1	S5/1	S1/2	S2/2	S3/2	S4/2
Aarau	1	1	1	1	1	1	1	2	2
Karlsruhe	4	4	4	4	4	4	4	4	4
Lehigh	3	3	3	3	3	3	3	3	3
Cardiff	2	2	2	2	2	2	2	1	1

	S1/1	S2/1	S3/1	S4/1	S5/1	S1/2	S2/2	S3/2	S4/2
Aarau	1	1	1	1	1	1	1	2	2
Karlsruhe	4	4	4	4	4	4	4	4	4
Lehigh	3	3	3	3	3	3	3	3	3
Cardiff	2	2	2	2	2	2	2	1	1

Note that Karlsruhe has the largest rank each time, indicating that its mean effect is greatest.

```
> round(apply(data_sub,1,mean),2)
```

S1/1	S2/1	S3/1	S4/1	S5/1	S1/2	S2/2	S3/2	S4/2
1.01	0.96	1.02	1.01	0.97	1.11	1.02	1.09	1.07

```
> round(apply(data_sub,2,mean),2)
```

Aarau	Karlsruhe	Lehigh	Cardiff
0.80	1.34	1.07	0.91

	S1/1	S2/1	S3/1	S4/1	S5/1	S1/2	S2/2	S3/2	S4/2
Aarau	1	1	1	1	1	1	1	2	2
Karlsruhe	4	4	4	4	4	4	4	4	4
Lehigh	3	3	3	3	3	3	3	3	3
Cardiff	2	2	2	2	2	2	2	1	1

```
>model.ranks =lm(c(data.ranks)~treat)
>anova(model.ranks)
```

Analysis of Variance Table

Response: c(data.ranks)

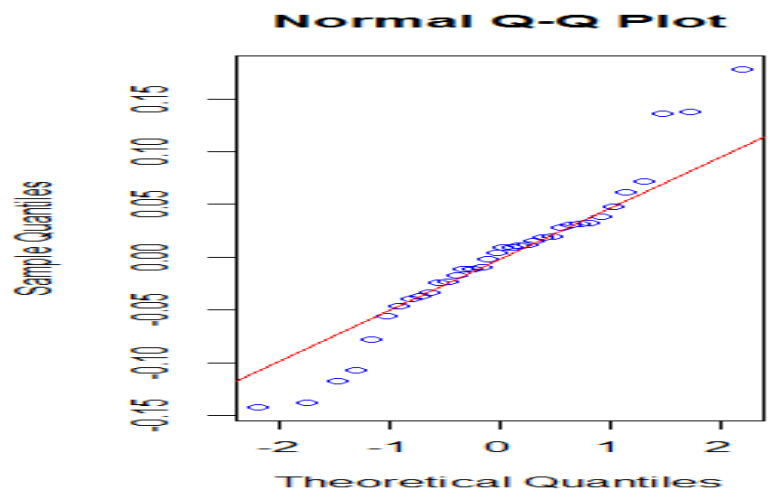
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treat	3	41.889	13.9630	143.62	< 2.2e-16 ***
Residuals	32	3.111	0.0972		

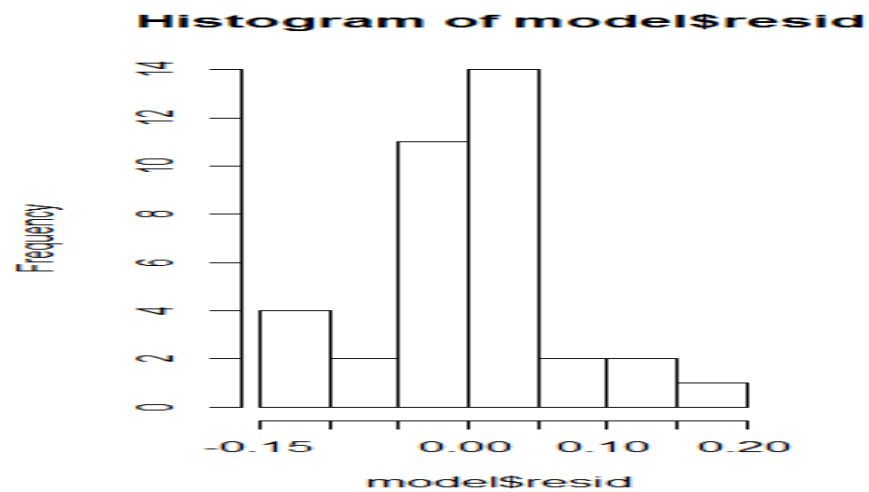
The usual ANOVA F-test, as before, does not take account of the fact that SS_{Total} is not random; the Friedman's test will do so. The differences are generally slight.

R Code to Perform Friedman's Test :

```
> friedman.test(y, treat,block)

        Friedman rank sum test
data:  y, treat and block
Friedman chi-squared = 25.1333, df = 3, p-value = 1.448e-05
```

The histogram is a fairly crude density estimator.





A much better estimator is the **kernel density estimator** (KDE). The estimator takes its name from so-called kernel function, denoted by K, which is probability density function that is symmetric about zero.

The standard normal density function is a common choice for K.

The **kernel density estimator** based on observations $y_1, y_2, \ldots, y_n$ is:

$$\hat{f}(y) = \frac{1}{nb}\sum_{i=1}^{n} K\left(\frac{y_i - y}{b}\right)$$

where b, which is called bandwidth, determine the resolution of the estimator.

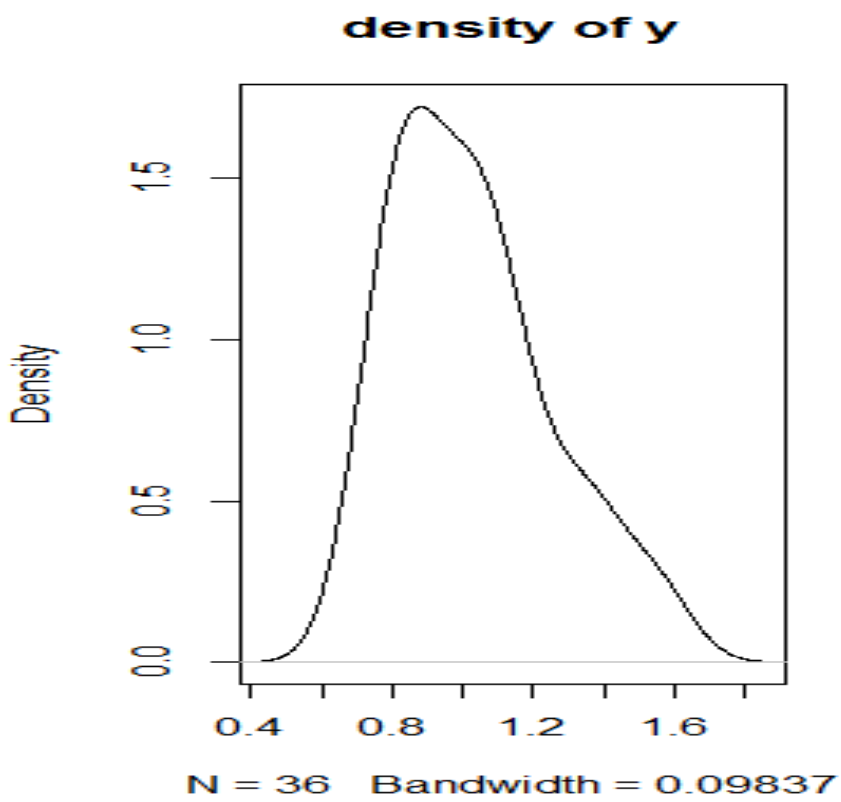
Choosing b requires to make a tradeoff between bias and variance. Appropriate values of b depend on both the sample size and the true density, of course, the latter is unknown, though it can be estimated.

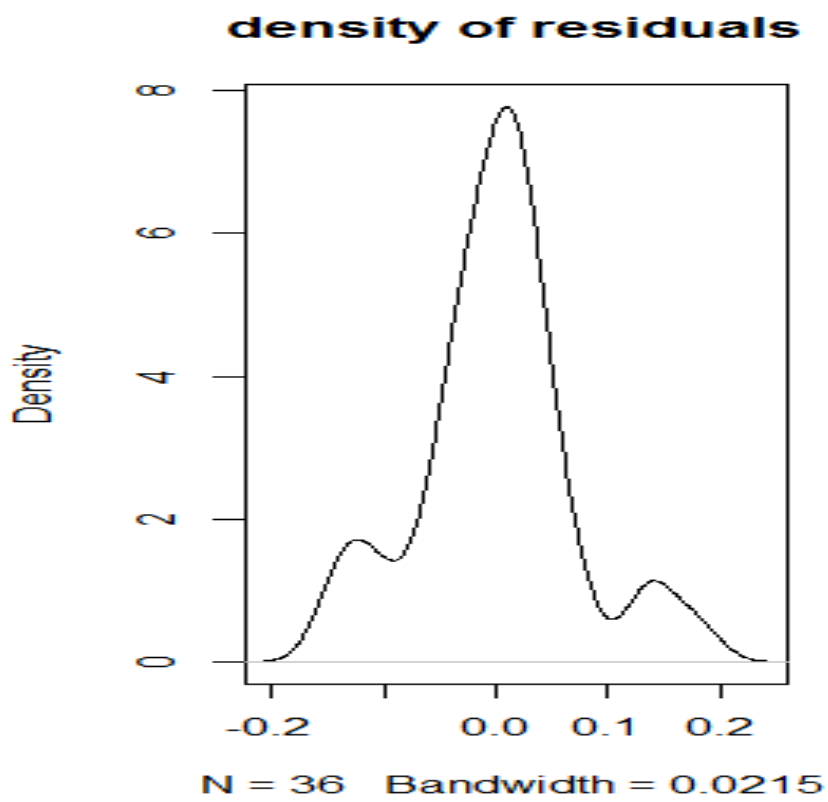
Fortunately, a large amount of research has been devoted to automatic selection of b.

Function density() computes kernel density estimates.

R code to find KDE:

```
> plot(density(y), main="density of y")
> plot(density(model$resid), main="density of residuals")
```





Box-Cox Transformation: There are a number reasons why data analysts often work, not with the original variables, but rather with transformation of the variables such as logs, square roots or other transformations.

The logarithm transformation is probably the most widely used transformation in data analysis, though the square root is a close choice. The log transformation is sometimes embedded into power transformation family by using Box-Cox transformation:

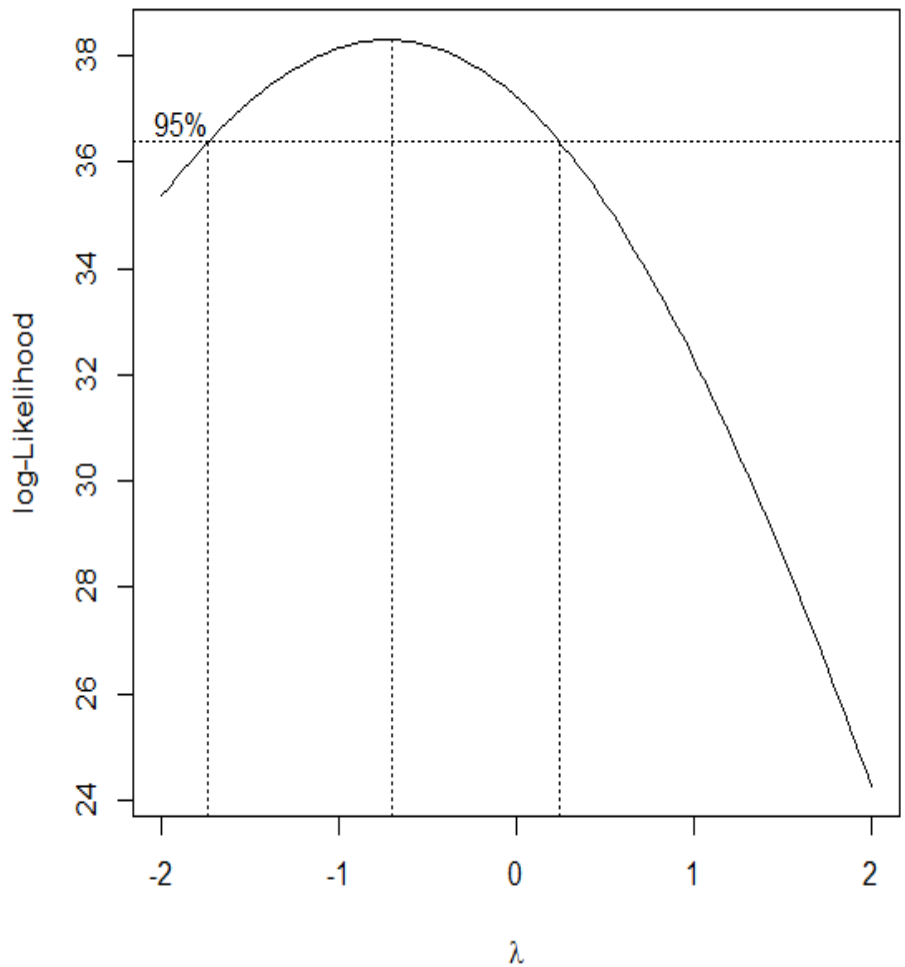
$$y^{(\alpha)} = \begin{cases} \dfrac{y^{\alpha} - 1}{\alpha}, & \alpha \neq 0 \\ Ln(y), & \alpha = 0 \end{cases}$$

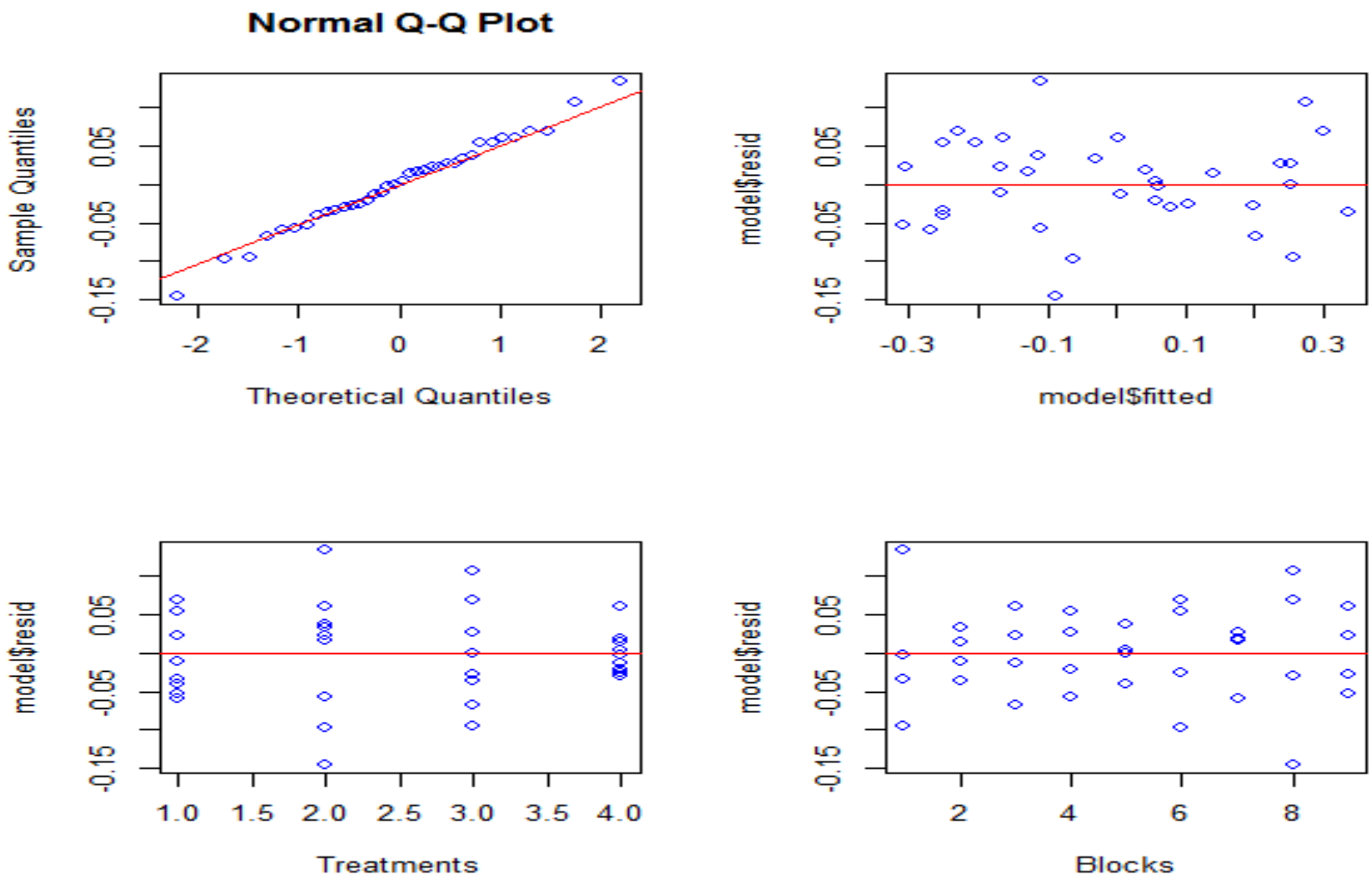
One way to determine the most appropriate value of α to use in the Box-Cox transformation is to plot the maximum likelihood function versus the value of used in transforming data. The value of that maximizes the log likelihood would be the most appropriate. This plot is called a Box-Cox plot.

The function **boxcox** function in the R package **MASS** makes Box-Cox plot automatically.

```
> library(MASS)
> bc = boxcox(model)
> alpha = bc$x[which.max(bc$y)]
> alpha
[1] -0.7070707
```

```
> ty=(y^(alpha)-1)/alpha
> model = lm(ty~ block+treat)
```




Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
model$resid
model$fitted
model$resid
Treatments
model$resid
Blocks

Suppose the assumptions were satisfied and now we would like to know which treatments are different from others. So, we can use individual or multiple comparison tests.

Fisher LSD: A confidence interval for **one** difference $\mu_i - \mu_j = \tau_i - \tau_j$, $i \neq j$, is

$$\bar{y}_{i.} - \bar{y}_{j.} \pm t_{\frac{\alpha}{2}, df(MSE)} \sqrt{MS_E} \sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$$

Assuming a two-sided alternative, the pair of means μ_i and μ_j would be declared significantly different if

$$\left|\bar{y}_{i.} - \bar{y}_{j.}\right| > t_{\frac{\alpha}{2}, df(MSE)} \sqrt{MS_E} \sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$$

The quantity $\mathrm{LSD} = t_{\frac{\alpha}{2}, df(MSE)} \sqrt{MS_E} \sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$ is called **least significant difference**.

Example 1:

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
block	8	0.08497	0.01062	1.5164	0.2036
treat	3	1.49997	0.49999	71.3830	4.211e-12 ***
Residuals	24	0.16810	0.00700		

$$\text{LSD} = t_{\frac{\alpha}{2},df(MSE)}\sqrt{MS_E}\sqrt{\frac{1}{n_i}+\frac{1}{n_j}} = t_{0.025,24}\sqrt{0.007}\sqrt{\frac{1}{9}+\frac{1}{9}}$$

$$= 2.064\sqrt{0.007}\sqrt{\frac{1}{9}+\frac{1}{9}} = 0.081$$

Therefore the null hypothesis is rejected if

$$\left|\bar{y}_{i.} - \bar{y}_{j.}\right| > 0.081$$

```
> round(apply(data_sub,2,mean),3)
```

Aarau	Karlsruhe	Lehigh	Cardiff
0.798	**1.340**	**1.066**	**0.906**

The null hypothesis is rejected if

$$|\bar{y}_{i.} - \bar{y}_{j.}| > 0.081$$

	Aarau	Karlsruhe	Lehigh
Karlsruhe	1.340-0.798=**0.542**		
Lehigh	1.066-0.798=**0.268**	1.066-1.340=**-0.274**	
Cardiff	0.906-0.798=**0.108**	0.906-1.340=**-0.434**	0.906-1.066=**-0.160**

μ_{Aarau} μ_{Cardiff} μ_{Lehigh} $\mu_{\text{Karlsruhe}}$

Tukey's Method:

$$k_\alpha = \frac{q_\alpha}{\sqrt{2}} = \frac{\text{qtukey}(0.95, \mathbf{4}, 24)}{\sqrt{2}} = 2.759$$

$$S.E.(\bar{y}_{i.} - \bar{y}_{j.}) = \sqrt{MS_{Error}}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}} \quad \text{for } i \neq j,\ i,j = 1,2,3,4$$

$$= \sqrt{0.007}\sqrt{\frac{1}{9} + \frac{1}{9}} = 0.039$$

The null hypothesis is rejected if

$$|\bar{y}_{i.} - \bar{y}_{j.}| > 2.759(0.039) = 0.108$$

μ_{Aarau} μ_{Cardiff} μ_{Lehigh} $\mu_{\text{Karlsruhe}}$

R Output for Tukey's Method:

```
> mtest = glht(model,linfct=mcp(treat="Tukey"))
> confint(mtest)
        Simultaneous Confidence Intervals
Multiple Comparisons of Means: Tukey Contrasts
Fit: lm(formula = y ~ block + treat)
Quantile = 2.761
95% family-wise confidence level
 Linear Hypotheses:
                          Estimate     lwr          upr
Cardiff - Aarau == 0      0.107333    -0.001597    0.216263
Karlsruhe - Aarau == 0    0.541889     0.432959    0.650819
Lehigh - Aarau == 0       0.268000     0.159070    0.376930
Karlsruhe - Cardiff == 0  0.434556     0.325626    0.543485
Lehigh - Cardiff == 0     0.160667     0.051737    0.269597
Lehigh - Karlsruhe == 0  -0.273889    -0.382819   -0.164959
```

Suppose that we **erroneously** analyzed these data as a CRD.

Question: What would the ANOVA be?

Answer: Recall that we have

$$SS_{Total} = \sum_{i=1}^{k} \sum_{j=1}^{b} (y_{ij} - \bar{y}_{..})^2$$

$$SS_{Total} = SS_{Treatments} + SS_{Error} \quad \text{for CRD}$$

and

$$SS_{Total} = SS_{Treatments} + SS_{Blocks} + SS_{Error} \quad \text{for RCBD}$$

In addition SS_{Total} is fixed for given data.

Therefore, the SS_{Blocks} must still be accounted for, and if it can't be anywhere else it ends up in experimental error.

```
> model <- lm(y~ treat+block)
> anova(model)
Analysis of Variance Table
Response: y
          Df  Sum Sq   Mean Sq   F value   Pr(>F)
treat      3  1.49997   0.49999   71.3830   4.211e-12 ***
block      8  0.08497   0.01062    1.5164   0.2036
Residuals 24  0.16810   0.00700
```

```
> model <- lm(y~treat)
> anova(model)
Analysis of Variance Table
Response: y
          Df  Sum Sq   Mean Sq   F value   Pr(>F)
treat      3  1.49997   0.49999   63.221    1.528e-13 ***
Residuals 32  0.25307   0.00791
```

$$0.08497 + 0.16810 = 0.25307$$

Example 2: The blood pressure of five patients were recorded before and 30 minutes after receiving a drug. The results are given in the following table.

Patients	1	2	3	4	5	Mean	S.D.
Before	146	134	110	126	157	134.6	18.105
After	140	138	110	116	144	129.6	13.822

Is the drug useful in **reducing** the blood pressure?

T-test for paired Samples:

```
> B = c(146, 134, 110, 126,157)
> A = c(140, 138, 110, 116, 144)
> t.test(B, A, alternative =  "greater", paired =T)

 Paired t-test

data:  B and A
t = 1.5972, df = 4, p-value = 0.09273
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 -1.673736       Inf
sample estimates:
mean of the differences
                      5
```

F-test for paired Samples:

```
> B = c(146, 134, 110, 126,157)
> A = c(140, 138, 110, 116, 144)
> y=c(B,A)
> Treat = factor(rep(c("B","A"), each=5))
> Block = factor(rep(c(1:5), 2))
> model=lm(y~Treat+Block)
> anova(model)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Treat	**1**	**62.5**	**62.5**	**2.551**	**0.185462**
Block	**4**	**2168.4**	**542.1**	**22.127**	**0.005447 ****
Residuals	**4**	**98.0**	**24.5**		

Latin Square: Sometimes there is **more** than one source of disturbance that can be eliminated by blocking.

Example 3: The following experiment was to test the feasibility of reducing air pollution by modifying a gasoline mixture with very small amount of certain chemicals, A, B, C, and D.

These four treatments were tested with four different drivers (I, II, III, and IV) and four different cars (1, 2, 3, and 4).

Therefore, there were **two block factors** – cars and drivers.

The **Latin square** design, shown in the next page, was used to help eliminate from the treatment comparisons possible differences between drivers and between cars. As you see each treatment A,B, C or D appears once in every row (driver) and once in every column (car).

Cars

Drivers	1	2	3	4
I	A 19	B 24	C 23	D 26
II	B 23	A 24	D 19	C 30
III	C 15	D 14	A 15	B 16
IV	D 19	C 18	B 19	A 16

Question: Is it OK to make 16 experimental runs with a single car and a single driver for the four different treatments?

Answer: Such design could also be statistically valid but Latin square design has the advantages that, it provides a wider basis for the conclusion drawn – any finding would not apply to one car and one driver.

R Codes to Create a Latin Square Design:

The function **design.lsd** in the R package “agricolae” can be used to create a Latin square design.

```
> library(agricolae)

> treat = c("A", "B", "C", "D")

> outdesign = design.lsd(treat, seed = 23)

> lsd = outdesign$book

> levels(lsd$row) = c("I", "II", "III", "IV")

> levels(lsd$col) = c(1,2,3,4)
```

Statistical Model: For $i, j, k = 1,2, \ldots, \mathrm{p}$,let

Response Variable

$$y_{ijk} = \mu + r_i + c_j + \tau_k + \varepsilon_{ijk}$$

Overall Mean

Row Effect

Column Effect

Treatment Effect

Random Error

where y_{ijk} is the observation in row i, column j, using treatment k.

Note: Note in example 3, $y_{133} = 23$, but y_{224} does not exist. There are only p^2 observations.

Note: The model is **additive** since there is no interaction effect. i.e. any treatment has the same effect regardless of the level of other factors.

Statistical Model: For $i, j, k = 1,2, \ldots, \mathrm{p}$, let

$$y_{ijk} = \mu + r_i + c_j + \tau_k + \varepsilon_{ijk}$$

Estimates:

- $\hat{\mu} = \bar{y}_{...}$
- $\hat{r}_i = \bar{y}_{i..}$-$\bar{y}_{...}$ for $i = 1,2, \ldots, p$
- $\hat{c}_j = \bar{y}_{.j.}$-$\bar{y}_{...}$ for j= $1,2, \ldots, p$
- $\tau_k = \bar{y}_{..k}$-$\bar{y}_{...}$ for k= $1,2, \ldots, p$

Then Residuals are:

$$e_{ijk} = y_{ijk} - \bar{y}_{i..} - \bar{y}_{.j.} - \bar{y}_{..k} + 2\bar{y}_{...}$$

Statistical Model: For $i, j, k = 1,2, \ldots, \mathrm{p}$, let

$$y_{ijk} = \mu + r_i + c_j + \tau_k + \varepsilon_{ijk}$$

Define

$$SS_{Total} = \sum_{i=1}^{p} \sum_{j=1}^{p} \sum_{k=1}^{p} (y_{ijk} - \bar{y}_{...})^2 ,$$

Note that the sumations are over observed (or exist) y_{ijk} .Similarly define:

$$SS_{Rows} = p \sum_{i=1}^{p} (\bar{y}_{i..} - \bar{y}_{...})^2 ,$$

$$SS_{Columns} = p \sum_{i=1}^{p} (\bar{y}_{.j.} - \bar{y}_{...})^2 ,$$

$$SS_{Treatments} = p \sum_{i=1}^{p} (\bar{y}_{..k} - \bar{y}_{...})^2 ,$$

Statistical Model: For $i, j, k = 1,2, \ldots, \mathrm{p}$, let

$$y_{ijk} = \mu + r_i + c_j + \tau_k + \varepsilon_{ijk}$$

It can be shown that

$$SS_{Total} = SS_{Treatments} + SS_{Rows} + SS_{Columns} + SS_{Error}$$

The degrees of freedom for testing $H_0: \tau_1 = \tau_2 = \cdots = \tau_p = 0$ (there is no treatment effect) are:

$$Extra\ df = p - 1$$

$$\begin{aligned} df(MS_{Error}) &= n - Number\ of\ parameters\ in\ the\ full\ model \\ &= p^2 - (1 + (p-1) + (p-1) + (p-1)\ ) \\ &= p^2 - 3p + 2 \\ &= (p-1)(p-2) \end{aligned}$$

The **ANOVA table**:

Source	df	SS	MS	F
Treatments	$p-1$	SSTr	$MSTr = \frac{SSTr}{p-1}$	$F = \frac{MSTr}{MSE}$
Rows	$p-1$	SSR	$MSR = \frac{SSR}{p-1}$	$F = \frac{MSR}{MSE}$
Columns	$p-1$	SSC	$MSC = \frac{SSC}{p-1}$	$F = \frac{MSC}{MSE}$
Error	$(p-1)(p-2)$	SSE	$MSE = \frac{SSE}{(p-1)(p-2)}$	
Total	$n-1$	SST		

R Codes to Define Factors in LSD:

Cars

Drivers	1	2	3	4
I	A 19	B 24	C 23	D 26
II	B 23	A 24	D 19	C 30
III	C 15	D 14	A 15	B 16
IV	D 19	C 18	B 19	A 16

```
> y  = c(19, 24, 23, 26, 23, 24, 19, 30,15, 14, 15, 16, 19, 18, 19, 16)
> drivers <- as.factor(rep(as.character(as.roman(1:4)),  each=4))
> cars<- as.factor(rep(1:4, times=4))
 > treat <- as.factor(c("A", "B", “C", “D", “B", “A", “D", “C",
        “C", "D", “A", “B",  “D", “C", "B", “A"))
```

Fuctions **head()** and **tails** can be used to return the **first** or **last** parts of a vector, matrix, table, data frame or function

```
> data <- data.frame(y, drivers, cars, treat)
>  head(data,10)

  1  19      I   1    A
  2  24      I   2    B
  3  23      I   3    C
  4  26      I   4    D
  5  23     II   1    B
  6  24     II   2    A
  7  19     II   3    D
  8  30     II   4    C
  9  15    III   1    C
 10  14    III   2    D
```

R Codes to Obtain ANOVA Table for LSD:

```
> model <- lm(y~treat+drivers+cars)
 > anova(model)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treat	3	20	6.667	0.7692	0.55181
drivers	3	216	72.000	8.3077	0.01477 *
cars	3	24	8.000	0.9231	0.48465
Residuals	6	52	8.667		

Signif. codes: 0 ‘*’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1**

The usual model checks should be done. In fact,

- To check **normality**: Use **qq-plot** of residuals.

- To check for **constant variances**: Use plots of residuals against
    a) row labels,
    b) column labels,
    c) treatment labels

You may also use **Bartlett's test** to check constant variances if normality is assured.

You may also use **Levene's test** to check constant variances.

R Codes to Perform Levene's test :

For Levene's test, we should compute the absolute deviations from the medians in each label group and then do a one-way ANOVA on them.

You may write a function to do Levene's test. This following is an example.

```
> my.levene.test = function(y, v, factor)
  {  y.d <- vector()
     for(j in v)  y.d[factor==j] = abs(y[factor==j]-median(y[factor==j]))
     model.d <- lm(y.d ~ factor)
     anova(model.d)
}
```

R Codes to Perform Levene's Test for Treatment Group:

```
> v = c("A", "B", "C", "D")
> my.levene.test(y, v, treat)
```

Analysis of Variance Table

Response: y.d

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor	3	12	4.0000	0.5106	0.6825
Residuals	12	94	7.8333		

When ANOVA F-test indicates that the means are different. You may use multiple comparisons methods to test which treatments are in fact differ.

```
> mtest <- glht(model,linfct=mcp(treat="Tukey"))
> confint(mtest)
```

Simultaneous Confidence Intervals

Multiple Comparisons of Means: Tukey Contrasts

Fit: lm(formula = y ~ treat + drivers + cars)

Quantile = 3.4634 95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
B - A == 0	2.0000	-5.2096	9.2096
C - A == 0	3.0000	-4.2096	10.2096
D - A == 0	1.0000	-6.2096	8.2096
C - B == 0	1.0000	-6.2096	8.2096
D - B == 0	-1.0000	-8.2096	6.2096
D - C == 0	-2.0000	-9.2096	5.2096

The Latin square notations extends to **Graeco – Latin squares** which is a $p \times p$ pattern that permits the study of p treatment simultaneously with three different blocking variables each at p levels.

Example 3 (Cont.): The $p \times p$ Graeco – Latin square shown in the next page is an extension of the Latin square design used earlier but with one extra blocking variables added.

The extra blocking variable is labeled $\alpha, \beta, \gamma, \delta$ and it could be used to eliminate possible difference between, say, four days on which the trials were run. Suppose, for example,

$\alpha = Monday, \beta = Tuesday$, $\gamma = Wednesday$, and $\delta = Thursday$

Note: Each (Latin, Greek) combinations of letters occurs exactly once.

Cars

Drivers	1	2	3	4
I	Aα 19	Bβ 24	Cγ 23	Dδ 26
II	Bδ 23	Aγ 24	Dβ 19	Cα 30
III	Cβ 15	Dα 14	Aδ 15	Bγ 16
IV	Dγ 19	Cδ 18	Bα 19	Aβ 16

$\alpha = Monday, \beta = Tuesday$, $\gamma = Wednesday$, and $\delta = Thursday$

The function **design.graeco** in the R package "agricolae" can be used to create a Graceo -Latin square design.

$$df(MS_{Error}) = n - Number\ of\ parameters\ in\ the\ full\ model$$

$$= p^2 - (1 + 4(p-1)\ ) = p^2 - 4p + 3$$

$$= (p-1)(p-3)$$

R Codes to Obtain ANOVA Table for Graeco – Latin squares:

```
> days <- as.factor(c("M","T","W", "TR", "TR","W","T","M",
                "T","M","TR","W", "W","TR","M","T"))
> model <- lm(y~treat+drivers+cars+days)
> anova(model)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treat	3	20	6.667	0.5	0.70821
drivers	3	216	72.000	5.4	0.09979 .
cars	3	24	8.000	0.6	0.65748
days	3	12	4.000	0.3	0.82542
Residuals	3	40	13.333		

Signif. codes: 0 ‘*’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1**

Randomized block designs are used to group heterogeneous experimental units together in homogeneous subgroups called blocks.

The number of experimental units per block is equal to the number of levels of the treatment factor in randomized complete block designs .

When then **number** of levels of treatment factor is **large** in a randomized block design, the corresponding number of experimental units per block must be large. This could cause a problem.

One solution to keep the number of experimental units per block small is to construct block designs where each block only contains a subset of the possible levels of treatment factor. Such designs are called **incomplete block designs**.

Example 4: Consider an experiment (Davies, O. L. ed. (1954), *The design and Analysis of Industrial Experiments*, London: Oliver and Boyd, page 200) to access the effect of tires made out of different compound on the lifetime of the tires.

Because of a manufacturing limitation, a tire can be divided into at **most three section**, with each section being made out of a different compound. Consequently, **only three compounds can be compared on the same unit** (i.e. tire).

As a particular tire is tested for resistance to wear, the three compounds on the tire are subjected to the same road conditions. Therefore, each tire is treated as a block. The wear data are reported as relative wear in arbitrary units.

The **balanced incomplete block design** (**BIBD**) used in the experiment as well as the wear data are given in table on the next pages

Example 4: **Compound**

Tire	A	B	C	D
1	238	238	279	
2	196	213		308
3	254		334	367
4		312	421	412

An incomplete block design is '**balanced**' if any **two treatments** appear in the **same block** an **equal** number of times. Therefore, the above design is a **Balanced Incomplete Block Design** (**BIBD**).

This is a symmetric design since $t = b$

$$\lambda = 2$$

Note:

There are $b = 4$ blocks of size $k = 3$.

There are $t = 4$ treatments and each treatment is replicated $r = 3$ times.

The number of times each treatment level occurs with every other treatment level in the same block is 2.

It can be shown that:

$$bk = rt \text{ and } r(k-1) = \lambda(t-1)$$

The above relations can be used to find the minimum number of blocks required for a BIBD.

Since r and λ must be integer, by second above equation, we see that $\lambda(t-1)$ must be divisible by $k-1$.

For example, if $t = 6$ and $k = 3$, then

$$r(k-1) = \lambda(t-1) \Rightarrow r(3-1) = \lambda(6-1) \Rightarrow 2r = 5\lambda$$

$$\Rightarrow 5\lambda \text{ must be divisible by } 2.$$

Therefore, the smallest integer for λ is $\lambda = 2$,

$r = \frac{10}{2} = 5$, and $b = \frac{rt}{k} = \frac{5\times 6}{3} = 10$.

The function **BIBsize** in R package **daewr** (design and analysis of experiments with R) provides a quick way of finding values of λ and r to satisfies $bk = rt$ and $r(k-1) = \lambda(t-1)$ when given the number of levels of treatment factor t and the block size k.

For example, in the code below the function is called with $t = 6$ and $k = 3$.

```
> install.packages("daewr")
> library(daewr)
> BIBsize(6,3)
```

Posible BIB design with b= 10 and r= 5 lambda= 2

$$bk = rt \text{ and } r(k-1) = \lambda(t-1)$$

It can be shown that:

- Even though the above equations are satisfied for some combinations of t, b, r, k and λ , a corresponding BIBD may not exists.
- If a BIBD does exist, it is a D-optimal design.

The function **optBlock** in the R package **AlgDesign** (Algorithmic Experimental Design) can be used to find a BIBD.

For example, suppose we would like to find a BIBD with $b = 10$ blocks of $k = 3$ experimental units per block and $t = 6$ levels of treatment factor. In the function optBlock use:

- option **blocksizes = rep(3,10**) to specify 10 block of size 3.
- option **withinData = factor(1:6)** specify six levels of one factor.

```
> library(AlgDesign)
> BIB = optBlock(~., withinData = factor(1:6),
  blocksizes = rep(3,10))
```

The object BIB created by **optBlock** function contains a data frame **BIB$design** with one column that contains a list of treatment factor levels for each of 10 blocks in order.

```
> des = BIB$rows
```

Note that creating the design again with the **optBlock** function may result in a different but equivalent BIB.

```
> dim(des) = NULL
> des = matrix(des, nrow=10, ncol=3, byrow=T,
  dimnames = list(paste(rep("Block",10), 1:10),
                  c("Unit 1", "Unit 2", "Unit 3")))
```

Statistical Model: For $i = 1,2,\ldots,t$ and $j = 1,2,\ldots,b$, let

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij},$$

Note that due to **incompleteness** of blocks $\bar{y}_{i.} - \bar{y}_{..}$ is no longer an unbiased estiamtes of τ_i.

For example 4, we have

$$E(\bar{y}_{2.}) = E\left(\frac{y_{21} + y_{22} + y_{24}}{3}\right) = \frac{1}{3}[Ey_{21} + Ey_{22} + Ey_{24}]$$

$$= \frac{1}{3}[(\mu + \tau_2 + \beta_1) + (\mu + \tau_2 + \beta_2) + (\mu + \tau_2 + \beta_4)]$$

$$= \mu + \tau_2 + \frac{\beta_1 + \beta_2 + \beta_4}{3}$$

Similarly, we have

$$E(\bar{y}_{..}) = \mu + \frac{r}{12}(\tau_1 + \tau_2 + \tau_3 + \tau_4) + \frac{k}{12}(\beta_1 + \beta_2 + \beta_3 + \beta_4) = \mu$$

Therefore, $E(\bar{y}_{2.} - \bar{y}_{..}) = \tau_2 + \frac{\beta_1+\beta_2+\beta_4}{3}$.

The "**incompleteness**" of the BIBD requires that a treatment total be adjusted not by the overall average because a treatment does not appear in every block.

The **adjusted treatment total** for the ith treatment can be expressed as:

$$Q_i = y_{i.} - \sum_{j=1}^{b} n_{ij}\bar{y}_{.j} = y_{i.} - \frac{1}{k}\sum_{j=1}^{b} n_{ij}y_{.j}$$

where $n_{ij} = 1$ if treatment i appears in block j and $n_{ij} = 0$ otherwise.

Note:

- $\sum_{j=1}^{b} n_{ij}y_{.j}$ is the total of the block total, counting only those blocks that contain treatment i.
- It can be shown that the **least squares estimate** $\hat{\tau}_i$ for τ_i are independent and the estimate and its variance are:

$$\hat{\tau}_i = \frac{k}{\lambda t}Q_i \text{ and } Var(\hat{\tau}_i) = \frac{k}{\lambda t}\sigma^2$$

Example 4: Compound

Tire	A	B	C	D	Total
1	238	238	279		755
2	196	213		308	717
3	254		334	367	955
4		312	421	412	1145
Total	688	763	1034	1087	3572

$$Q_i = y_{i.} - \sum_{j=1}^{b} n_{ij}\bar{y}_{.j} = y_{i.} - \frac{1}{k}\sum_{j=1}^{b} n_{ij}y_{.j}$$

$$Q_1 = 688 - \frac{1}{3}(755 + 717 + 955) = -121$$

$$Q_2 = 763 - \frac{1}{3}(755 + 717 + 1145) = -109.\bar{3}$$

$$Q_3 = 1034 - \frac{1}{3}(755 + 955 + 1145) = 82.\bar{3}$$

$$Q_4 = 1087 - \frac{1}{3}(717 + 955 + 1145) = 148$$

Note: It is always the case that

$$\sum_{i=1}^{b} Q_i = 0$$

Statistical Model: For $i = 1,2,\ldots,t$ and $j = 1,2,\ldots,b$, let

$$y_{ij} = \mu + \beta_j + \tau_i + \varepsilon_{ij}$$

Define

$$SS_{Total} = \sum_{i=1}^{t}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{..})^2 ,$$

where the summation is over all observed (i, j).

$$SS_{Blocks} = k\sum_{j=1}^{b}(\bar{y}_{.j} - \bar{y}_{..})^2 ,$$

$$SS_{Treatments(Adjusted)} = \frac{k}{\lambda t}\sum_{i=1}^{t} Q_i^2$$

$$SS_{Error} = SS_{Total} - SS_{Blocks} - SS_{Treatments(adjusted)}$$

The idea is that we first estimate the block effects and then see how much of the remaining variation is attributable to treatments. Doing it in the other order results in something quite different.

The **ANOVA table**:

Source	df	SS	MS	F
Blocks	$\mathrm{b}-1$	SSB	$\mathrm{MSTr}=\dfrac{\mathrm{SS}B}{\mathrm{b}-1}$	
Treatments (adjusted)	$\mathrm{t}-1$	SSTr(adj)	$\mathrm{MSB}=\dfrac{\mathrm{SS}Tr(adj)}{\mathrm{t}-1}$	$\mathrm{F}=\dfrac{\mathrm{MS}Tr(adj)}{\mathrm{MSE}}$
Error	$\mathrm{b}k-\mathrm{b}-\mathrm{t}+1$	SSE	$\mathrm{MSE}=\dfrac{\mathrm{SSE}}{bk-b-t-1}$	
Total	$\mathrm{bk}-1$	SST		

R Codes to Obtain ANOVA Table for BIBD:

```
> y = c(238,238,279, 196,213,308, 254, 334, 367,312,421,412)
> treat = as.factor(c("A", "B", "C","A", "B", "D",
                      "A", "C", "D","B", "C", "D"))
> block = as.factor(rep(1:4, each=3))
> model <- lm(y ~ block + treat)
> anova(model)
```

For a BIBD, ANOVA depends on the order of the treatment and block factors entering the model. In the above, the treatment effects are adjusted for the block effects

The following ANOVA table is NOT right.

```
> model2 <- lm(y~ treat + block)
> anova(model2)
```

ANOVA Table for BIBD:

```
> model <- lm(y ~ block + treat)
> anova(model)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
block	3	39123	13040.9	37.240	0.0007618 ***
treat	3	20729	6909.7	19.732	0.0033516 **
Residuals	5	1751	350.2		

Since there is a treatment effect, we can now use multiple comparisons methods. Note:

$$S.E.(\hat{\tau}_i - \hat{\tau}_j) = \sqrt{\frac{2k}{\lambda t} MS_{Error}}$$

What Have We Learned?

Blocking is an **extremely important** design technique, used **extensively** in industrial experimentation.

We discussed blocking in this chapter. In particular, we discussed following designs:

- Randomized Complete Block Designs (RCBD)
- Latin Square Designs (LSD)
- Graeco – Latin squares
- Balanced Incomplete Block Designs (BIBD)